



北京大學

第四讲 朴素贝叶斯

机器学习概论

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- 基本概念
- 主要模型
 - 伯努利模型
 - 类型模型
 - 高斯模型
 - 各模型参数最大似然估计
- 预测
 - 最大类标签后验
- 贝叶斯-朴素贝叶斯
 - 参数后验分布
 - 预测
 - 参数最大后验估计

- 生成模型

$$p(y = c|x; \theta) = \frac{p(y = c; \theta)p(x|y = c; \theta)}{p(x; \theta)} = \frac{p(y = c; \theta)p(x|y = c; \theta)}{\sum_{i=1}^C p(y = i; \theta)p(x|y = i; \theta)}$$

- $p(y = c; \theta)$: 类标签先验分布

- $p(x|y = c; \theta)$: 数据 x 的类条件分布; 通过采样 $p(x|y = c; \theta)$, 能够生成类别 c 的数据 x

- θ : 模型参数

- 朴素贝叶斯假定

- 记 $\mathbf{x} = [x_1, x_2, \dots, x_D]^T$

- 给定类别标签 $c \in \{1, \dots, C\}$, 数据的特征相互独立, 即

$$p(\mathbf{x}|y = c; \theta) = \prod_{d=1}^D p(x_d|y = c; \theta)$$

- 类标签后验概率

$$p(y = c | \mathbf{x}; \boldsymbol{\theta}) = \frac{p(y = c; \boldsymbol{\theta}) \prod_{d=1}^D p(x_d | y = c; \boldsymbol{\theta})}{\sum_{i=1}^C p(y = i; \boldsymbol{\theta}) \prod_{d=1}^D p(x_d | y = i; \boldsymbol{\theta})}$$

– 记 $\alpha_c = p(y = c; \boldsymbol{\theta})$, $\beta_{dc} = p(x_d | y = c; \boldsymbol{\theta})$, 则 $\boldsymbol{\theta} = \{\alpha_c\} \cup \{\beta_{dc}\}$ 是模型的所有参数

- 以上模型称为朴素贝叶斯 (Naïve Bayes)

– 简单, 尽管类条件独立要求过强, 但实践中很有效

– 在文本分类、垃圾邮件过滤、推荐系统和医疗诊断等有广泛的应用

- $\mathbf{x}$ 是 D 维离散数据, 第 i 维有 K_i 个取值

- 类条件独立

$$p(\mathbf{x} | y = c; \boldsymbol{\theta}) = \prod_{d=1}^D p(x_d | y = c; \boldsymbol{\theta})$$

– 给定类标签 c , 类条件概率数量 $\sum_{i=1}^D K_i$

- 无类条件独立

$$p(\mathbf{x} | y = c; \boldsymbol{\theta}) = p(x_1, \dots, x_D | y = c; \boldsymbol{\theta})$$

– 给定类标签 c , 类条件概率数量 $\prod_{i=1}^D K_i$

- $x_d \in \{0,1\}$, 特征类条件分布可用伯努利分布建模

$$-p(\mathbf{x}|y = c; \boldsymbol{\theta}) = \prod_{d=1}^D \beta_{dc}^{x_d} (1 - \beta_{dc})^{1-x_d} = \prod_{d=1}^D \text{Ber}(x_d; \beta_{dc})$$

- $x_d \in \{1, 2, \dots, K_d\}$, 特征类条件分布可用类型分布建模

$$-p(\mathbf{x}|y = c; \boldsymbol{\theta}) = \prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\mathbb{I}(x_d=k)} = \prod_{d=1}^D \text{Cat}(x_d; \{\beta_{kdc}\})$$

- $x_d \in \mathbb{R}$, 特征类条件分布可用高斯分布建模

$$-p(\mathbf{x}|y = c; \boldsymbol{\theta}) = \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dc}, \sigma_{dc}^2)$$

模型参数

如何学习各类模型的参数?

- 给定训练数据集 $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$
- 模型参数 $\boldsymbol{\theta} = \{\alpha_c\} \cup \{\beta_{dc}\}$
 - $\alpha_c = p(y = c), \beta_{dc} = p(x_d = 1|y = c)$
- 目标：求解 $\hat{\boldsymbol{\theta}}$ ，使得

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\mathcal{D}; \boldsymbol{\theta})$$

- 即，模型参数配置为 $\boldsymbol{\theta}^*$ 时，训练样本集 $\mathcal{D}$ 出现的概率最大
- 方法
 - 最大似然估计 (MLE)

$$\begin{aligned}
 p(\mathcal{D}; \boldsymbol{\theta}) &= p(\{(\mathbf{x}_n, y_n)\}_{n=1}^N; \boldsymbol{\theta}) = \prod_{n=1}^N p(\mathbf{x}_n, y_n; \boldsymbol{\theta}) \\
 &= \prod_{n=1}^N (p(y_n; \boldsymbol{\theta}) p(\mathbf{x}_n | y_n; \boldsymbol{\theta})) = \prod_{n=1}^N p(y_n; \boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{x}_n | y_n; \boldsymbol{\theta}) \\
 &= \left(\prod_{n=1}^N \prod_{c=1}^C (p(c; \boldsymbol{\theta}))^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C (p(\mathbf{x}_n | c; \boldsymbol{\theta}))^{\mathbb{I}(y_n=c)} \right) \\
 &\quad \begin{array}{l} \boldsymbol{\theta} = \{\alpha_c\} \cup \{\beta_{dc}\} \\ \alpha_c = p(y=c), \quad \beta_{dc} = p(x_d=1|y=c) \end{array} \quad p(\mathbf{x} | y=c; \boldsymbol{\theta}) = \prod_{d=1}^D \beta_{dc}^{x_{nd}} (1 - \beta_{dc})^{1-x_{nd}} \\
 &= \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \beta_{dc}^{x_{nd}} (1 - \beta_{dc})^{1-x_{nd}} \right)^{\mathbb{I}(y_n=c)} \right)
 \end{aligned}$$

$$p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \beta_{dc}^{x_{nd}} (1 - \beta_{dc})^{1-x_{nd}} \right)^{\mathbb{I}(y_n=c)} \right)$$



$$\text{LL}(\boldsymbol{\theta}) = \log p(\mathcal{D}; \boldsymbol{\theta})$$

$$= \sum_{n=1}^N \sum_{c=1}^C \mathbb{I}(y_n = c) \log \alpha_c + \sum_{n=1}^N \sum_{c=1}^C \mathbb{I}(y_n = c) \left(\sum_{d=1}^D (x_{nd} \log \beta_{dc} + (1 - x_{nd}) \log(1 - \beta_{dc})) \right)$$

$$= \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{n=1}^N \mathbb{I}(y_n = c) (x_{nd} \log \beta_{dc} + (1 - x_{nd}) \log(1 - \beta_{dc}))$$

最大似然估计: $\max_{\theta} \text{LL}(\theta)$ 约束条件: $\sum_{c=1}^C \alpha_c = 1$

$$\text{LL}(\theta) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{n=1}^N \mathbb{I}(y_n = c) (x_{nd} \log \beta_{dc} + (1 - x_{nd}) \log(1 - \beta_{dc}))$$


独立子优化问题: $\theta_C = \{\alpha_c\}$

$C \times D$ 个独立子优化问题: β_{dc}

$$\max_{\theta_C} \text{LL}_C(\theta_C) \quad \text{约束条件: } \sum_{c=1}^C \alpha_c = 1$$

$$\text{LL}_C(\theta_C) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c$$

$$\max_{\beta_{dc}} \text{LL}_{dc}(\beta_{dc})$$

$$\text{LL}_{dc}(\beta_{dc}) = \sum_{n=1}^N \mathbb{I}(y_n = c) (x_{nd} \log \beta_{dc} + (1 - x_{nd}) \log(1 - \beta_{dc}))$$

$$\max_{\boldsymbol{\theta}_C} \text{LL}_C(\boldsymbol{\theta}_C) \quad \text{约束条件: } \sum_{c=1}^C \alpha_c = 1$$

$$\text{LL}_C(\boldsymbol{\theta}_C) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c$$

带等式约束极值问题，经典求解方法：拉格朗日乘子法

$$\mathcal{L}(\boldsymbol{\theta}_C, \lambda) = \text{LL}_C(\boldsymbol{\theta}_C) + \lambda \left(1 - \sum_{c=1}^C \alpha_c \right) \quad \text{求解} \begin{cases} \frac{\partial \mathcal{L}(\boldsymbol{\theta}_C, \lambda)}{\partial \lambda} = 0 \\ \frac{\partial \mathcal{L}(\boldsymbol{\theta}_C, \lambda)}{\partial \boldsymbol{\theta}_C} = 0 \end{cases} \xrightarrow{\boldsymbol{\theta}_C = \{\alpha_i\}} \frac{\partial \mathcal{L}(\boldsymbol{\theta}_C, \lambda)}{\partial \alpha_i} = 0; 1 \leq i \leq C$$

$$\mathcal{L}(\boldsymbol{\theta}_C, \lambda) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \lambda \left(1 - \sum_{c=1}^C \alpha_c \right)$$

$\alpha_i = p(y = i)$ 是类标签 c 的概率

类标签为 i 的样本数量

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}_C, \lambda)}{\partial \lambda} = 0 \quad \longrightarrow \quad 1 - \sum_{c=1}^C \alpha_c = 0$$

$$\sum_{c=1}^C \alpha_c = 1$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}_C, \lambda)}{\partial \alpha_i} = 0$$

$$\frac{1}{\alpha_i} \sum_{n=1}^N \mathbb{I}(y_n = i) - \lambda = 0 \quad \longrightarrow \quad \alpha_i = \frac{1}{\lambda} \sum_{n=1}^N \mathbb{I}(y_n = i)$$

$$\lambda = \sum_{i=1}^C \sum_{n=1}^N \mathbb{I}(y_n = i) = \sum_{n=1}^N \sum_{i=1}^C \mathbb{I}(y_n = i) = N$$

$$\hat{\alpha}_i = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(y_n = i)$$

$$\max_{\beta_{dc}} \text{LL}_{dc}(\beta_{dc})$$

$$\text{LL}_{dc}(\beta_{dc}) = \sum_{n=1}^N \mathbb{I}(y_n = c) (x_{nd} \log \beta_{dc} + (1 - x_{nd}) \log(1 - \beta_{dc}))$$

可微分函数极大值在该函数导数或梯度上的值为零

可微分严格上凸函数最大值在其导数或梯度上的值为零，反之亦然

$$\frac{\partial \text{LL}_{dc}(\beta_{dc})}{\partial \beta_{dc}} = 0 \rightarrow \sum_{n=1}^N \mathbb{I}(y_n = c) \left(\frac{x_{nd}}{\beta_{dc}} - \frac{1 - x_{nd}}{1 - \beta_{dc}} \right) = 0$$

第 d 个维度取值为1的 c 类样本数量

$$\beta_{dc} = \frac{\sum_{n=1}^N \mathbb{I}(y_n = c) x_{nd}}{\sum_{n=1}^N \mathbb{I}(y_n = c)} = \frac{\sum_{n=1}^N \mathbb{I}(y_n = c) x_{nd}}{N_c}$$

$$\beta_{dc} = p(x_d = 1 | y = c)$$

类标签为 c 的样本数量 $N_c \triangleq \sum_{n=1}^N \mathbb{I}(y_n = c)$

给定类别 c ，第 d 个维度取值为1的概率

- 给定训练数据集 $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$
- 模型参数 $\boldsymbol{\theta} = \{\alpha_c\} \cup \{\beta_{kdc}\}$
 - $\alpha_c = p(y = c), \quad \beta_{kdc} = p(x_d = k | y = c)$
- 目标：求解 $\hat{\boldsymbol{\theta}}$ ，使得

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\mathcal{D}; \boldsymbol{\theta})$$

- 即，模型参数配置为 $\boldsymbol{\theta}^*$ 时，训练样本集 $\mathcal{D}$ 出现的概率最大
- 方法
 - 最大似然估计

$$\begin{aligned}
 p(\mathcal{D}; \boldsymbol{\theta}) &= p(\{(\mathbf{x}_n, y_n)\}_{n=1}^N; \boldsymbol{\theta}) = \prod_{n=1}^N p(\mathbf{x}_n, y_n; \boldsymbol{\theta}) \\
 &= \prod_{n=1}^N (p(y_n; \boldsymbol{\theta}) p(\mathbf{x}_n | y_n; \boldsymbol{\theta})) = \prod_{n=1}^N p(y_n; \boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{x}_n | y_n; \boldsymbol{\theta}) \\
 &= \left(\prod_{n=1}^N \prod_{c=1}^C (p(c; \boldsymbol{\theta}))^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C (p(\mathbf{x}_n | c; \boldsymbol{\theta}))^{\mathbb{I}(y_n=c)} \right) \\
 &\quad \alpha_c = p(y = c), \beta_{kdc} = p(x_d = k | y = c) \quad p(\mathbf{x} | y = c; \boldsymbol{\theta}) = \prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\mathbb{I}(x_d=k)} \\
 &= \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\mathbb{I}(x_{n,d}=k)} \right)^{\mathbb{I}(y_n=c)} \right)
 \end{aligned}$$

$$p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\mathbb{I}(x_{n,d}=k)} \right)^{\mathbb{I}(y_n=c)} \right)$$



$$\text{LL}(\boldsymbol{\theta}) = \log p(\mathcal{D}; \boldsymbol{\theta}) = \sum_{n=1}^N \sum_{c=1}^C \mathbb{I}(y_n = c) \log \alpha_c + \sum_{n=1}^N \sum_{c=1}^C \mathbb{I}(y_n = c) \left(\sum_{d=1}^D \sum_{k=1}^{K_d} \mathbb{I}(x_{n,d} = k) \log \beta_{kdc} \right)$$



$$\text{LL}(\boldsymbol{\theta}) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{k=1}^{K_d} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = k) \log \beta_{kdc}$$

最大似然估计: $\max_{\boldsymbol{\theta}} \text{LL}(\boldsymbol{\theta})$ 约束条件: $\begin{cases} \sum_{c=1}^C \alpha_c = 1 \\ \sum_{k=1}^{K_d} \beta_{kdc} = 1; 1 \leq d \leq D, 1 \leq c \leq C \end{cases}$

求解方法同
伯努利模型

$$\text{LL}(\boldsymbol{\theta}) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{k=1}^{K_d} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = k) \log \beta_{kdc}$$

独立子优化问题: $\boldsymbol{\theta}_C = \{\alpha_c\}$

$C \times D$ 个独立子优化问题: $\boldsymbol{\theta}_{dc} = \{\beta_{kdc}\}$

$$\max_{\boldsymbol{\theta}_C} \text{LL}_C(\boldsymbol{\theta}_C) \quad \text{约束条件: } \sum_{c=1}^C \alpha_c = 1$$

$$\text{LL}_C(\boldsymbol{\theta}_C) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c$$

$$\max_{\boldsymbol{\theta}_{dc}} \text{LL}_{dc}(\boldsymbol{\theta}_{dc}) \quad \text{约束条件: } \sum_{k=1}^{K_d} \beta_{kdc} = 1$$

$$\text{LL}_{dc}(\boldsymbol{\theta}_{dc}) = \sum_{k=1}^{K_d} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = k) \log \beta_{kdc}$$

$$\max_{\boldsymbol{\theta}_{dc}} \text{LL}_{dc}(\boldsymbol{\theta}_{dc}) \quad \text{约束条件: } \sum_{k=1}^{K_d} \beta_{kdc} = 1$$

$$\text{LL}_{dc}(\boldsymbol{\theta}_{dc}) = \sum_{k=1}^{K_d} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = k) \log \beta_{kdc}$$

拉格朗日乘子法

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda) \\ = \text{LL}_{dc}(\boldsymbol{\theta}_{dc}) + \lambda \left(1 - \sum_{k=1}^{K_d} \beta_{kdc} \right) \end{aligned} \quad \text{求解} \left\{ \begin{array}{l} \frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \lambda} = 0 \\ \frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \boldsymbol{\theta}_{dc}} = 0 \end{array} \right. \xrightarrow{\boldsymbol{\theta}_{dc} = \{\beta_{idc}\}} \frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \beta_{idc}} = 0; 1 \leq i \leq K_d$$

$$\mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda) = \sum_{k=1}^{K_d} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = k) \log \beta_{kdc} + \lambda \left(1 - \sum_{k=1}^{K_d} \beta_{kdc} \right) \quad N_{idc} \triangleq \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = i)$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \lambda} = 0 \quad \longrightarrow \quad 1 - \sum_{k=1}^{K_d} \beta_{kdc} = 0$$

$$\sum_{k=1}^{K_d} \beta_{kdc} = 1$$

类标签为 c 且第 d 维值为 i 的样本数量

$$\hat{\beta}_{idc} = \frac{\sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = i)}{N_c} = \frac{N_{idc}}{N_c}$$

?

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \beta_{idc}} = 0$$

$$\frac{1}{\beta_{idc}} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = i) - \lambda = 0$$

$$\beta_{idc} = \frac{1}{\lambda} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = i)$$

$$\lambda = \sum_{i=1}^{K_d} \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{n,d} = i) = \sum_{n=1}^N \mathbb{I}(y_n = c) = N_c$$

$\beta_{idc} = p(x_d = i | y = c)$ 是类标签 c 的条件下, 第 d 维值为 i 的概率

- 给定训练数据集 $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$
- 模型参数 $\boldsymbol{\theta} = \{\alpha_c\} \cup \{\beta_{kdc}\}$
 - $\alpha_c = p(y = c)$, $p(\mathbf{x} | y = c; \boldsymbol{\theta}) = \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dc}, \sigma_{dc}^2)$
- 目标: 求解 $\hat{\boldsymbol{\theta}}$, 使得

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\mathcal{D}; \boldsymbol{\theta})$$

- 即, 模型参数配置为 $\hat{\boldsymbol{\theta}}$ 时, 训练样本集 $\mathcal{D}$ 出现的概率最大
- 方法
 - 最大似然估计

$$\begin{aligned} p(\mathcal{D}; \boldsymbol{\theta}) &= p(\{(\mathbf{x}_n, y_n)\}_{n=1}^N; \boldsymbol{\theta}) = \prod_{n=1}^N p(\mathbf{x}_n, y_n; \boldsymbol{\theta}) \\ &= \prod_{n=1}^N (p(y_n; \boldsymbol{\theta}) p(\mathbf{x}_n | y_n; \boldsymbol{\theta})) = \prod_{n=1}^N p(y_n; \boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{x}_n | y_n; \boldsymbol{\theta}) \\ &= \left(\prod_{n=1}^N \prod_{c=1}^C (p(c; \boldsymbol{\theta}))^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C (p(\mathbf{x}_n | y_n; \boldsymbol{\theta}))^{\mathbb{I}(y_n=c)} \right) \\ &\quad \downarrow \begin{array}{l} \alpha_c = p(y=c) \quad p(\mathbf{x}_n | y=c; \boldsymbol{\theta}) = \prod_{d=1}^D \mathcal{N}(x_{n,d}; \mu_{dc}, \sigma_{dc}^2) \end{array} \\ &= \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \mathcal{N}(x_{n,d}; \mu_{dc}, \sigma_{dc}^2) \right)^{\mathbb{I}(y_n=c)} \right) \end{aligned}$$

$$p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \mathcal{N}(x_{n,d}; \mu_{dc}, \sigma_{dc}^2) \right)^{\mathbb{I}(y_n=c)} \right)$$

$$\text{LL}(\boldsymbol{\theta}) = \log p(\mathcal{D}; \boldsymbol{\theta}) = \sum_{n=1}^N \sum_{c=1}^C \mathbb{I}(y_n = c) \log \alpha_c + \sum_{n=1}^N \sum_{c=1}^C \mathbb{I}(y_n = c) \left(\sum_{d=1}^D \left(-\frac{\log 2\pi}{2} - \frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right) \right)$$

$$\text{LL}(\boldsymbol{\theta}) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{\log 2\pi}{2} - \frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right)$$

$$\text{LL}(\boldsymbol{\theta}) \triangleq \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right)$$

最大似然估计: $\max_{\theta} \text{LL}(\theta)$ 约束条件: $\sum_{c=1}^C \alpha_c = 1$

$$\text{LL}(\theta) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c + \sum_{c=1}^C \sum_{d=1}^D \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right)$$

求解方法同
伯努利模型

独立子优化问题: $\theta_c = \{\alpha_c\}$

$\max_{\theta_c} \text{LL}_c(\theta_c)$ 约束条件: $\sum_{c=1}^C \alpha_c = 1$

$$\text{LL}_c(\theta_c) = \sum_{c=1}^C \sum_{n=1}^N \mathbb{I}(y_n = c) \log \alpha_c$$

$C \times D$ 个独立子优化问题: $\theta_{dc} = \{\mu_{dc}, \sigma_{dc}^2\}$

$\max_{\theta_{dc}} \text{LL}_{dc}(\theta_{dc})$

$$\text{LL}_{dc}(\theta_{dc}) = \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right)$$

$$\max_{\boldsymbol{\theta}_{dc}} \text{LL}_{dc}(\boldsymbol{\theta}_{dc})$$

$$\text{LL}_{dc}(\boldsymbol{\theta}_{dc}) = \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right)$$

$$\text{求解 } \frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \boldsymbol{\theta}_{dc}} = \mathbf{0} \xrightarrow{\boldsymbol{\theta}_{dc} = \{\mu_{dc}, \sigma_{dc}^2\}} \begin{cases} \frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \mu_{dc}} = 0 \\ \frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \sigma_{dc}^2} = 0 \end{cases}$$

类标签为 c 的样本在第 d 维上的均值

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \mu_{dc}} = 0 \xrightarrow{\quad} \sum_{n=1}^N \mathbb{I}(y_n = c) \left(2 \frac{x_{n,d} - \mu_{dc}}{2\sigma_{dc}^2} \right) = 0 \xrightarrow{\quad} \mu_{dc} = \frac{\sum_{n=1}^N \mathbb{I}(y_n = c) x_{n,d}}{N_c}$$

$$\text{LL}_{dc}(\boldsymbol{\theta}_{dc}) = \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{\log \sigma_{dc}^2}{2} - \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^2} \right)$$

$$\frac{\partial \mathcal{L}(\boldsymbol{\theta}_{dc}, \lambda)}{\partial \sigma_{dc}^2} = 0 \quad \longrightarrow \quad \sum_{n=1}^N \mathbb{I}(y_n = c) \left(-\frac{1}{2\sigma_{dc}^2} + \frac{(x_{n,d} - \mu_{dc})^2}{2\sigma_{dc}^4} \right) = 0$$

$$\sigma_{dc}^2 = \frac{\sum_{n=1}^N \mathbb{I}(y_n = c) (x_{n,d} - \mu_{dc})^2}{N_c}$$

类标签为 c 的样本在第 d 维上的方差

- 通过训练数据集 $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$, 学习到模型参数 θ 。则如下概率分布均确定 (可计算)

- $- p(y = c; \theta)$

- $- p(x_d | y = c; \theta)$

- 对类别未知样本 $\mathbf{x}$, 可通过以下公式计算其类标签后验概率

$$p(y = c | \mathbf{x}; \theta) = \frac{p(y = c; \theta) \prod_{d=1}^D p(x_d | y = c; \theta)}{\sum_{i=1}^C p(y = i; \theta) \prod_{d=1}^D p(x_d | y = i; \theta)}$$

- 预测 $\mathbf{x}$ 类标签 $\hat{c}$ 的方法

- 最大类标签后验

$$\hat{c} = \max_c p(c | \mathbf{x}; \hat{\theta})$$

- 为什么要用最大类标签后验来预测类标签
- 假定类标签 $c \in \{1, \dots, C\}$, 定义选择 c 作为 x 类标签的期望风险如下

$$R(c|x) = E_{p(a|x)} l(c, a) = \sum_{a=1}^C p(a|x) l(c, a)$$

– 其中 $l(c, a)$ 表示如果 x 的真实类标签为 a , 但预测成 c 带来的损失。上式定义选择 c 为 x 类标签所带来的风险为 **错误选择 c 带来的损失在类标签后验分布上的期望**

- 进一步, 假定 $l(c, a)$ 为 0-1 损失函数

$$l(c, a) = \begin{cases} 1, & c \neq a \\ 0, & c = a \end{cases}$$

– 则
$$R(c|x) = E_{p(a|x)} l(c, a) = \sum_{a=1}^C p(a|x) l(c, a) = \sum_{a \neq c} p(a|x) = 1 - p(c|x)$$

- 为此，我们选择风险最小的类标签作为 x 的类标签

$$\min_c R(c|x)$$

- 由 $R(c|x) = 1 - p(c|x)$, 知

$$\min_c R(c|x) = \min_c (1 - p(c|x)) = \max_c p(c|x)$$

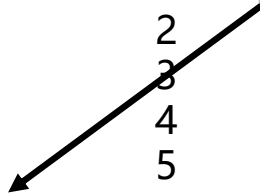
- 所以，选则最大后验的类标签具有最小的期望风险
- 以上分析基于贝叶斯决策理论 (Bayesian decision theory)
 - 务必注意： 上述结论在0-1损失函数下才成立
 - 所以，对于非0-1损失函数，通过最大后验选择的类标签就不一定是贝叶斯最优的（期望风险最小）

Table3.1 爬山气候记录

日期ID	天气	温度	湿度	风力	爬山
1	晴	热	高	弱	否
2	晴	热	高	强	否
3	阴	热	高	弱	是
4	雨	适中	高	弱	是
5	雨	冷	中	弱	是
6	雨	冷	中	强	否
7	阴	冷	中	强	是
8	晴	适中	中	弱	否
9	晴	冷	中	弱	是
10	雨	适中	中	弱	是
11	晴	适中	中	强	是
12	阴	适中	高	强	是
13	阴	热	中	弱	是
14	阴	冷	高	强	是
15	雨	适中	高	强	否


 标签

问：在“天气为雨，温度适中，湿度中和风力强的气候情况下”，是否适合爬山？


 天气数据

- 符号约定

- $F_{\text{天气}}$ 表示代表天气特征的随机变量
- $F_{\text{温度}}$ 表示代表温度特征的随机变量
- $F_{\text{湿度}}$ 表示代表湿度特征的随机变量
- $F_{\text{风力}}$ 表示代表风力特征的随机变量
- $D_{\text{爬山}}$ 表示代表是否爬山（决策）的随机变量

- 问题形式化

- 给定 $F_{\text{天气}} = \text{雨}$, $F_{\text{温度}} = \text{适中}$, $F_{\text{湿度}} = \text{中}$, $F_{\text{风力}} = \text{强}$ 的条件下, $D_{\text{爬山}} = \text{是}$ 和 否 的概率, 即

$$p(D_{\text{爬山}} = \text{是} | F_{\text{天气}} = \text{雨}, F_{\text{温度}} = \text{适中}, F_{\text{湿度}} = \text{中}, F_{\text{风力}} = \text{强})$$

$$p(D_{\text{爬山}} = \text{否} | F_{\text{天气}} = \text{雨}, F_{\text{温度}} = \text{适中}, F_{\text{湿度}} = \text{中}, F_{\text{风力}} = \text{强})$$

- 为讨论方便, 上述条件概率简记为
 - $p(\text{是} | \text{雨}, \text{适中}, \text{中}, \text{强})$
 - $p(\text{否} | \text{雨}, \text{适中}, \text{中}, \text{强})$
- 如果能计算出以上两个概率值, 我们就能做出决策
 - $p(\text{是} | \text{雨}, \text{适中}, \text{中}, \text{强}) > p(\text{否} | \text{雨}, \text{适中}, \text{中}, \text{强})$
 - ✓ 爬山
 - $p(\text{是} | \text{雨}, \text{适中}, \text{中}, \text{强}) < p(\text{否} | \text{雨}, \text{适中}, \text{中}, \text{强})$
 - ✓ 不爬山
- 方法
 - 采用朴素贝叶斯计算上述条件概率及做出预测

- 贝叶斯公式 $p(y|x) = \frac{p(x|y)p(y)}{p(x)}$

– 令 $x = (\text{雨}, \text{适中}, \text{中}, \text{强})$, $y = \text{是}$, 则

$$p(\text{是} | \text{雨}, \text{适中}, \text{中}, \text{强}) = \frac{p(\text{雨}, \text{适中}, \text{中}, \text{强} | \text{是})p(\text{是})}{p(\text{雨}, \text{适中}, \text{中}, \text{强})}$$

- 条件独立

$$p(\text{雨}, \text{适中}, \text{中}, \text{强} | \text{是}) = p(\text{雨} | \text{是}) p(\text{适中} | \text{是}) p(\text{中} | \text{是}) p(\text{强} | \text{是})$$

- 边际概率

$$p(x) = \sum_y p(x, y) = \sum_y p(x|y)p(y)$$

- 因此, 需要学习 (估计) 的概率分布 (参数)

– $p(D_{\text{爬山}})$

– $p(F_{\text{天气}} | \text{是})$, $p(F_{\text{温度}} | \text{是})$, $p(F_{\text{湿度}} | \text{是})$, $p(F_{\text{风力}} | \text{是})$

– $p(F_{\text{天气}} | \text{否})$, $p(F_{\text{温度}} | \text{否})$, $p(F_{\text{湿度}} | \text{否})$, $p(F_{\text{风力}} | \text{否})$

日期ID	天气	温度	湿度	风力	爬山
1	晴	热	高	弱	否
2	晴	热	高	强	否
3	阴	热	高	弱	是
4	雨	适中	高	弱	是
5	雨	冷	中	弱	是
6	雨	冷	中	强	否
7	阴	冷	中	强	是
8	晴	适中	高	弱	否
9	晴	冷	中	弱	是
10	雨	适中	中	弱	是
11	晴	适中	中	强	是
12	阴	适中	中	强	是
13	阴	热	高	弱	是
14	阴	冷	高	强	是
15	雨	适中	高	强	否

$$\hat{\alpha}_i = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(y_n = i) \rightarrow \begin{cases} p(D_{\text{爬山}} = \text{是}) = \frac{10}{15} = \frac{2}{3} \\ p(D_{\text{爬山}} = \text{否}) = \frac{5}{15} = \frac{1}{3} \end{cases}$$

$$\hat{\beta}_{idc} = \frac{\sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_d = i)}{N_c}$$

$$p(F_{\text{天气}} = \text{晴} | \text{是}) = \frac{2}{10} = \frac{1}{5}$$

$$p(F_{\text{天气}} = \text{阴} | \text{是}) = \frac{5}{10} = \frac{1}{2}$$

$$p(F_{\text{天气}} = \text{雨} | \text{是}) = \frac{3}{10}$$

- 先验概率 $p(D_{爬山})$
 - $p(D_{爬山})$: $p(是) = \frac{2}{3}$, $p(否) = \frac{1}{3}$
- 条件概率:
 - $p(F_{天气}|是)$: $p(晴|是) = \frac{1}{5}$, $p(阴|是) = \frac{1}{2}$, $p(雨|是) = \frac{3}{10}$
 - $p(F_{温度}|是)$: $p(热|是) = \frac{1}{5}$, $p(适中|是) = \frac{2}{5}$, $p(冷|是) = \frac{2}{5}$
 - $p(F_{湿度}|是)$: $p(高|是) = \frac{2}{5}$, $p(中|是) = \frac{3}{5}$
 - $p(F_{风力}|是)$: $p(强|是) = \frac{2}{5}$, $p(弱|是) = \frac{3}{5}$
 - $p(F_{天气}|否)$: $p(晴|否) = \frac{3}{5}$, $p(阴|否) = 0$, $p(雨|否) = \frac{2}{5}$
 - $p(F_{温度}|否)$: $p(热|否) = \frac{2}{5}$, $p(适中|否) = \frac{2}{5}$, $p(冷|否) = \frac{1}{5}$
 - $p(F_{湿度}|否)$: $p(高|否) = \frac{4}{5}$, $p(中|否) = \frac{1}{5}$
 - $p(F_{风力}|否)$: $p(强|否) = \frac{3}{5}$, $p(弱|否) = \frac{2}{5}$

问：在“天气为雨，温度适中，湿度中和风力强的气候情况下”，是否适合爬山？

$$p(\text{是}|\text{雨, 适中, 强}) = \frac{p(\text{雨, 适中, 中, 强}|\text{是})p(\text{是})}{p(\text{雨, 适中, 中, 强})} = \frac{p(\text{雨, 适中, 中, 强}|\text{是})p(\text{是})}{p(\text{雨, 适中, 中, 强}|\text{是})p(\text{是}) + p(\text{雨, 适中, 中, 强}|\text{否})p(\text{否})}$$

$$p(\text{是}|\text{雨, 适中, 中, 强}) = \frac{p(\text{雨}|\text{是})p(\text{适中}|\text{是})p(\text{中}|\text{是})p(\text{强}|\text{是})p(\text{是})}{p(\text{雨}|\text{是})p(\text{适中}|\text{是})p(\text{中}|\text{是})p(\text{强}|\text{是})p(\text{是}) + p(\text{雨}|\text{否})p(\text{适中}|\text{否})p(\text{中}|\text{否})p(\text{强}|\text{否})p(\text{否})}$$

$$p(\text{是}|\text{雨, 适中, 中, 强}) = \frac{\frac{3}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{2}{3}}{\frac{3}{10} \times \frac{2}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{2}{3} + \frac{2}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{3}{5} \times \frac{1}{3}} = \frac{3}{4}$$

$$\text{同理, } p(\text{否}|\text{雨, 适中, 中, 强}) = \frac{1}{4}$$

$\frac{3}{4} > \frac{1}{4}$ 因此, 建议爬山

问：在“天气为雨，温度适中，湿度中和风力强的气候情况下”，是否适合爬山？

$$p(\text{是}|\text{雨, 适中, 强}) = \frac{p(\text{雨, 适中, 中, 强}|\text{是})p(\text{是})}{p(\text{雨, 适中, 中, 强})} \propto p(\text{雨, 适中, 中, 强}|\text{是})p(\text{是})$$

$$p(\text{否}|\text{雨, 适中, 中, 强}) = \frac{p(\text{雨, 适中, 中, 强}|\text{否})p(\text{否})}{p(\text{雨, 适中, 中, 强})} \propto p(\text{雨, 适中, 中, 强}|\text{否})p(\text{否})$$

因此，对于分类任务而言

比较 $p(\text{雨, 适中, 中, 强}|\text{是})p(\text{是})$ 和 $p(\text{雨, 适中, 中, 强}|\text{否})p(\text{否})$ 大小即可

即，不需要计算出 $p(\text{雨, 适中, 中, 强})$

假定在湿度属性还有一个取值“低”。但由于爬山气候记录时长不够或其他原因，没有记录到湿度值为“低”的天气情况。通过朴素贝叶斯方法可知：

日期ID	天气	温度	湿度	风力	爬山
1	晴	热	高	弱	否
2	晴	热	高	强	否
3	阴	热	高	弱	是
4	雨	适中	高	弱	是
5	雨	冷	中	弱	是
6	雨	冷	中	强	否
7	阴	冷	中	强	是
8	晴	适中	中	弱	否
9	晴	冷	中	弱	是
10	雨	适中	中	弱	是
11	晴	适中	中	强	是
12	阴	适中	中	强	是
13	阴	热	中	弱	是
14	阴	冷	高	强	是
15	雨	适中	高	强	否

$$p(\text{低}|\text{是}) \triangleq p(\text{湿度} = \text{低}|\text{是}) = 0$$

$$p(\text{低}|\text{否}) \triangleq p(\text{湿度} = \text{低}|\text{否}) = 0$$

如果问：在“天气为雨，温度适中，湿度低和风力强的气候情况下”，是否适合爬山？

$$\begin{aligned}
 & p(\text{雨}, \text{适中}, \text{低}, \text{强}|\text{是})p(\text{是}) \\
 &= p(\text{雨}|\text{是})p(\text{适中}|\text{是})p(\text{低}|\text{是})p(\text{强}|\text{是})p(\text{是}) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 & p(\text{雨}, \text{适中}, \text{低}, \text{强}|\text{否})p(\text{否}) \\
 &= p(\text{雨}|\text{否})p(\text{适中}|\text{否})p(\text{低}|\text{否})p(\text{强}|\text{否})p(\text{否}) \\
 &= 0
 \end{aligned}$$

无法判断

另外由于训练数据的随机性造成出现零概率的估计
 $(p(\text{阴}|\text{否}) = 0)$ 也是不符合现实的

- 通过引入模型参数先验，解决数据缺失或数据分布不均的问题

- 以类型模型为例，介绍贝叶斯方法参数

- $\theta = \{\alpha, \beta\}, \alpha = \{\alpha_c\}, \beta = \{\beta_{dc}\}, \beta_{dc} = \{\beta_{kdc}\}$

- $\alpha_c = p(y = c), \beta_{kdc} = p(x_d = k | y = c)$

- 方法

- θ 不是未知数，而是服从一定分布的随机变量

- 通过训练样本集 $\mathcal{D}$ ，学习参数的后验分布 $p(\theta | \mathcal{D})$

- 基于参数后验分布，对类标签未知样本 x 的类标签分布 $p(y = c | x, \mathcal{D})$ 进行推断

- ✓ 贝叶斯估计（基于期望）

- ✓ 最大后验估计（点估计）

- $\alpha = \{\alpha_1, \dots, \alpha_c\}$ 服从以下狄利克雷分布

$$\alpha \sim \text{Dir}(\tilde{\alpha}) = \frac{1}{B(\tilde{\alpha})} \prod_{c=1}^C (\alpha_c)^{\tilde{\alpha}_c - 1} \quad \tilde{\alpha} = \{\tilde{\alpha}_1, \dots, \tilde{\alpha}_C\}, \tilde{\alpha}_c > 0$$

- $\beta_{dc} = \{\beta_{1dc}, \dots, \beta_{K_d dc}\}$ 服从以下狄利克雷分布

$$\beta_{dc} \sim \text{Dir}(\tilde{\beta}_{dc}) = \frac{1}{B(\tilde{\beta}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\tilde{\beta}_{kdc} - 1}$$

超参数，凭经验
人工设定

$$\tilde{\beta}_{dc} = \{\tilde{\beta}_{1dc}, \dots, \tilde{\beta}_{K_d dc}\}, \tilde{\beta}_{kdc} > 0$$

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}$$

$$p(\boldsymbol{\theta}) = p(\{\boldsymbol{\alpha}, \boldsymbol{\beta}\}) = p(\boldsymbol{\alpha})p(\boldsymbol{\beta}) = p(\boldsymbol{\alpha})p(\{\boldsymbol{\beta}_{dc}\}) = p(\boldsymbol{\alpha}) \prod_{c=1}^C \prod_{d=1}^D p(\boldsymbol{\beta}_{dc})$$

$$p(\boldsymbol{\alpha}) = \frac{1}{B(\tilde{\boldsymbol{\alpha}})} \prod_{c=1}^C (\alpha_c)^{\tilde{\alpha}_c - 1} \quad \downarrow \quad p(\boldsymbol{\beta}_{dc}) = \frac{1}{B(\tilde{\boldsymbol{\beta}}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\tilde{\beta}_{kdc} - 1}$$

$$p(\boldsymbol{\theta}) = \frac{1}{B(\tilde{\boldsymbol{\alpha}})} \prod_{c=1}^C (\alpha_c)^{\tilde{\alpha}_c - 1} \prod_{c=1}^C \prod_{d=1}^D \frac{1}{B(\tilde{\boldsymbol{\beta}}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\tilde{\beta}_{kdc} - 1}$$

$$p(\mathcal{D}|\boldsymbol{\theta}) = p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{n=1}^N \prod_{c=1}^C \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{n=1}^N \prod_{c=1}^C \left(\prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\mathbb{I}(x_{nd}=k)} \right)^{\mathbb{I}(y_n=c)} \right)$$



$$p(\mathcal{D}|\boldsymbol{\theta}) = p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{c=1}^C \prod_{n=1}^N \alpha_c^{\mathbb{I}(y_n=c)} \right) \left(\prod_{c=1}^C \prod_{d=1}^D \prod_{k=1}^{K_d} \prod_{n=1}^N \beta_{kdc}^{\mathbb{I}(x_{nd}=k)\mathbb{I}(y_n=c)} \right)$$



$$p(\mathcal{D}|\boldsymbol{\theta}) = p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{c=1}^C \alpha_c^{\sum_{n=1}^N \mathbb{I}(y_n=c)} \right) \left(\prod_{c=1}^C \prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\sum_{n=1}^N \mathbb{I}(y_n=c)\mathbb{I}(x_{nd}=k)} \right)$$

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}$$

$$p(\boldsymbol{\theta}) = \frac{1}{B(\tilde{\boldsymbol{\alpha}})} \prod_{c=1}^C (\alpha_c)^{\tilde{\alpha}_c-1} \prod_{c=1}^C \prod_{d=1}^D \frac{1}{B(\tilde{\boldsymbol{\beta}}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\tilde{\beta}_{kdc}-1}$$

$$p(\mathcal{D}|\boldsymbol{\theta}) = p(\mathcal{D}; \boldsymbol{\theta}) = \left(\prod_{c=1}^C \alpha_c^{\sum_{n=1}^N \mathbb{I}(y_n=c)} \right) \left(\prod_{c=1}^C \prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\sum_{n=1}^N \mathbb{I}(y_n=c) \mathbb{I}(x_{nd}=k)} \right)$$



$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{\left(\prod_{c=1}^C \alpha_c^{\sum_{n=1}^N \mathbb{I}(y_n=c) + \tilde{\alpha}_c - 1} \right) \left(\prod_{c=1}^C \prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\sum_{n=1}^N \mathbb{I}(y_n=c) \mathbb{I}(x_{nd}=k) + \tilde{\beta}_{kdc} - 1} \right)}{p(\mathcal{D}) B(\tilde{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D B(\tilde{\boldsymbol{\beta}}_{dc})}$$

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{\left(\prod_{c=1}^C \alpha_c^{\sum_{n=1}^N \mathbb{I}(y_n=c) + \tilde{\alpha}_c - 1}\right) \left(\prod_{c=1}^C \prod_{d=1}^D \prod_{k=1}^{K_d} \beta_{kdc}^{\sum_{n=1}^N \mathbb{I}(y_n=c) \mathbb{I}(x_{nd}=k) + \tilde{\beta}_{kdc} - 1}\right)}{p(\mathcal{D}) B(\tilde{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D B(\tilde{\boldsymbol{\beta}}_{dc})}$$

$$Z \triangleq p(\mathcal{D}) B(\tilde{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D B(\tilde{\boldsymbol{\beta}}_{dc}) \quad f(\boldsymbol{\alpha}) \triangleq \prod_{c=1}^C \alpha_c^{N_c + \tilde{\alpha}_c - 1} \quad N_c \triangleq \sum_{n=1}^N \mathbb{I}(y_n = c)$$

$$g_{cd}(\boldsymbol{\beta}_{dc}) \triangleq \prod_{k=1}^{K_d} \beta_{kdc}^{N_{kdc} + \tilde{\beta}_{kdc} - 1} \quad N_{kdc} \triangleq \sum_{n=1}^N \mathbb{I}(y_n = c) \mathbb{I}(x_{nd} = k)$$

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{1}{Z} f(\boldsymbol{\alpha}) \left(\prod_{c=1}^C \prod_{d=1}^D g_{cd}(\boldsymbol{\beta}_{dc}) \right)$$

$$f(\boldsymbol{\alpha}) \triangleq \prod_{c=1}^C \alpha_c^{N_c + \tilde{\alpha}_c - 1} \propto \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}}) = \frac{1}{B(\bar{\boldsymbol{\alpha}})} \prod_{c=1}^C (\alpha_c)^{\bar{\alpha}_c - 1} \quad \begin{aligned} \bar{\boldsymbol{\alpha}} &= \{\bar{\alpha}_1, \dots, \bar{\alpha}_C\} \\ \bar{\alpha}_c &= N_c + \tilde{\alpha}_c \end{aligned}$$

$$g_{cd}(\boldsymbol{\beta}_{dc}) \triangleq \prod_{k=1}^{K_d} \beta_{kdc}^{N_{kdc} + \tilde{\beta}_{kdc} - 1} \propto \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc}) = \frac{1}{B(\bar{\boldsymbol{\beta}}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\bar{\beta}_{kdc} - 1}$$

$$\bar{\boldsymbol{\beta}}_{dc} = \{\bar{\beta}_{1dc}, \dots, \bar{\beta}_{K_d dc}\} \quad \bar{\beta}_{K_d dc} = N_{kdc} + \tilde{\beta}_{kdc}$$

$$p(\boldsymbol{\theta} | \mathcal{D}) \propto \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})$$

$$\boldsymbol{\theta} = \{\boldsymbol{\alpha}\} \cup \{\boldsymbol{\beta}_{dc}\}$$



$$p(\boldsymbol{\theta}|\mathcal{D}) = p(\boldsymbol{\alpha}, \boldsymbol{\beta}_{11}, \dots, \boldsymbol{\beta}_{DC}|\mathcal{D}) = p(\boldsymbol{\alpha}|\mathcal{D}) \prod_{c=1}^C \prod_{d=1}^D p(\boldsymbol{\beta}_{dc}|\mathcal{D})$$

$$p(\boldsymbol{\theta}|\mathcal{D}) \propto \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})$$



$$p(\boldsymbol{\alpha}|\mathcal{D}) = \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}})$$

$$p(\boldsymbol{\beta}_{dc}|\mathcal{D}) = \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})$$

给定类别未知样本 $\mathbf{x}$, 预测其类别 $y^* = \underset{y}{\operatorname{argmax}} p(y|\mathbf{x}, \mathcal{D})$

$$p(y = c|\mathbf{x}, \mathcal{D}) = \frac{p(y = c|\mathcal{D})p(\mathbf{x}|y = c, \mathcal{D})}{p(\mathbf{x}|\mathcal{D})}$$

$$\propto p(y = c|\mathcal{D})p(\mathbf{x}|y = c, \mathcal{D})$$

$$= p(y = c|\mathcal{D}) \prod_{d=1}^D \prod_{k=1}^{K_d} p(x_d = k|y = c, \mathcal{D})^{\mathbb{I}(x_d=k)}$$

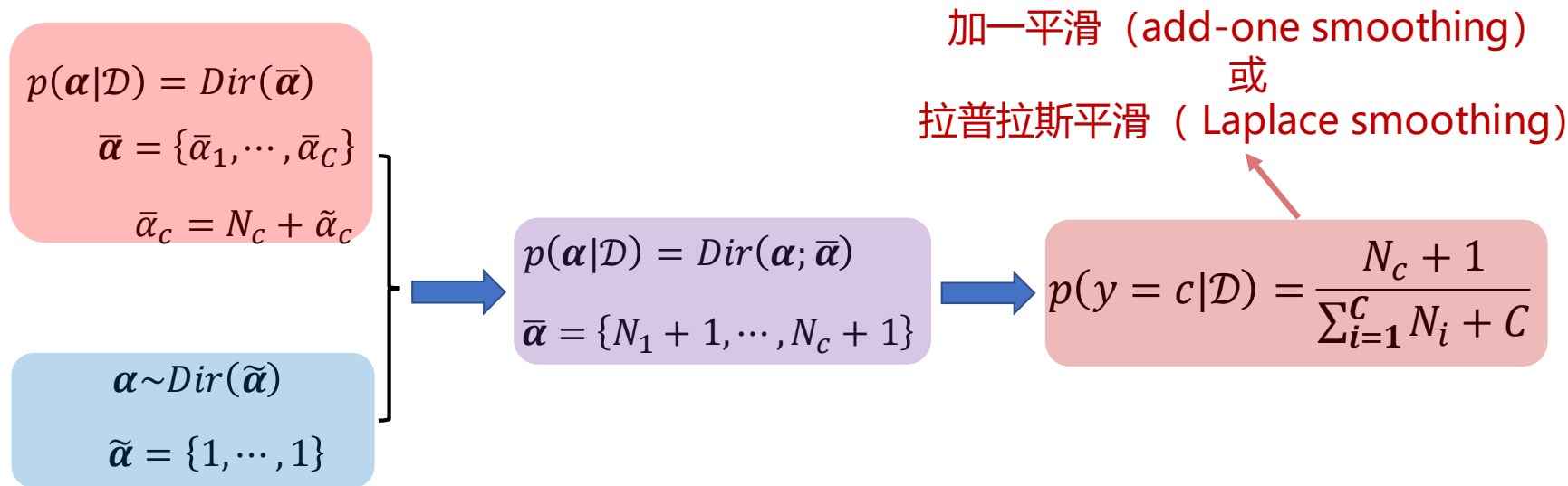
$$p(y = c|\mathcal{D})?$$

$$p(x_d = k|y = c, \mathcal{D})?$$

$$\begin{aligned}
 p(y = c|\mathcal{D}) &= \int p(y = c, \boldsymbol{\alpha}|\mathcal{D}) d\boldsymbol{\alpha} \\
 &= \int p(y = c|\boldsymbol{\alpha}, \mathcal{D})p(\boldsymbol{\alpha}|\mathcal{D}) d\boldsymbol{\alpha} \\
 &= \int p(y = c|\boldsymbol{\alpha})p(\boldsymbol{\alpha}|\mathcal{D}) d\boldsymbol{\alpha} \\
 \boldsymbol{\alpha} = \{\alpha_1, \dots, \alpha_c\} &\downarrow \alpha_c = p(y = c|\boldsymbol{\alpha}) \\
 &= \int \alpha_c p(\boldsymbol{\alpha}|\mathcal{D}) d\boldsymbol{\alpha} \xrightarrow{p(\boldsymbol{\alpha}|\mathcal{D}) = \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}})} \int \alpha_c \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}}) d\boldsymbol{\alpha} \\
 &\quad \uparrow \bar{\boldsymbol{\alpha}} = \{\bar{\alpha}_1, \dots, \bar{\alpha}_c\} \quad \bar{\alpha}_c = N_c + \tilde{\alpha}_c \\
 &\quad = \mathbb{E}_{\boldsymbol{\alpha} \sim \text{Dir}(\bar{\boldsymbol{\alpha}})}[\alpha_c] \\
 &\quad \uparrow \\
 &\quad = \frac{N_c + \tilde{\alpha}_c}{\sum_{i=1}^c (N_i + \tilde{\alpha}_i)} \\
 &\quad \sum_{i=1}^c N_i = N \quad \uparrow \\
 &\quad = \frac{N_c + \tilde{\alpha}_c}{N + \sum_{i=1}^c \tilde{\alpha}_i}
 \end{aligned}$$

$$\begin{aligned}
 p(x_d = k | y = c, \mathcal{D}) &= \int p(x_d = k, \boldsymbol{\beta}_{dc} | y = c, \mathcal{D}) d\boldsymbol{\beta}_{dc} \\
 &= \int p(x_d = k | \boldsymbol{\beta}_{dc}, y = c, D) p(\boldsymbol{\beta}_{dc} | y = c, D) d\boldsymbol{\beta}_{dc} \\
 &= \int p(x_d = k | \boldsymbol{\beta}_{dc}, y = c) p(\boldsymbol{\beta}_{dc} | D) d\boldsymbol{\beta}_{dc} \\
 \boldsymbol{\beta}_{dc} = \{\beta_{1dc}, \dots, \beta_{K_d dc}\} &\downarrow \beta_{kdc} = p(x_d = k | y = c) \\
 &= \int \beta_{kdc} p(\boldsymbol{\beta}_{dc} | D) d\boldsymbol{\beta}_{dc} \xrightarrow{p(\boldsymbol{\beta}_{dc} | D) = \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})} = \int \beta_{kdc} \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc}) d\boldsymbol{\beta}_{dc} \\
 &= \mathbb{E}_{\boldsymbol{\beta}_{dc} \sim \text{Dir}(\bar{\boldsymbol{\beta}}_{dc})} [\beta_{kdc}] \\
 \bar{\boldsymbol{\beta}}_{dc} = \{\bar{\beta}_{1dc}, \dots, \bar{\beta}_{K_d dc}\} &\uparrow \bar{\beta}_{K_d dc} = N_{kdc} + \tilde{\beta}_{kdc} \\
 &= \frac{N_{kdc} + \tilde{\beta}_{kdc}}{N_c + \sum_{i=1}^{K_d} \tilde{\beta}_{idc}} \\
 N_c = \sum_{i=1}^{K_d} N_{idc} &\uparrow N_{kdc} + \tilde{\beta}_{kdc} \\
 &= \frac{N_{kdc} + \tilde{\beta}_{kdc}}{\sum_{i=1}^{K_d} (N_{idc} + \tilde{\beta}_{idc})}
 \end{aligned}$$

$$\begin{aligned}
 p(y = c | \mathbf{x}, \mathcal{D}) &\propto p(\mathbf{x} | y = c, \mathcal{D}) \prod_{d=1}^D \prod_{k=1}^{K_d} p(x_d = k | y = c, \mathcal{D})^{\mathbb{I}(x_d=k)} \\
 p(\mathbf{x} | y = c, \mathcal{D}) &= \frac{N_c + \tilde{\alpha}_c}{\sum_{i=1}^c (N_i + \tilde{\alpha}_i)} \quad \downarrow \quad p(x_d = k | y = c, \mathcal{D}) = \frac{N_{kdc} + \tilde{\beta}_{kdc}}{N_c + \sum_{i=1}^{K_d} \tilde{\beta}_{idc}} \\
 p(y = c | \mathbf{x}, \mathcal{D}) &\propto \frac{N_c + \tilde{\alpha}_c}{\sum_{i=1}^c (N_i + \tilde{\alpha}_i)} \prod_{d=1}^D \prod_{k=1}^{K_d} \left(\frac{N_{kdc} + \tilde{\beta}_{kdc}}{N_c + \sum_{i=1}^{K_d} \tilde{\beta}_{idc}} \right)^{\mathbb{I}(x_d=k)}
 \end{aligned}$$



$$\alpha \sim \text{Dir}(\tilde{\alpha}) = \frac{1}{B(\tilde{\alpha})} \prod_{c=1}^C (\alpha_c)^{\tilde{\alpha}_c - 1} = \frac{1}{B(\{1, \dots, 1\})} \longrightarrow \alpha \text{ 的先验分布是均匀分布}$$

$$p(\boldsymbol{\beta}_{dc}|\mathcal{D}) = \text{Dir}(\bar{\boldsymbol{\beta}}_{dc})$$

$$\bar{\boldsymbol{\beta}}_{dc} = \{\bar{\beta}_{1dc}, \dots, \bar{\beta}_{K_d dc}\}$$

$$\bar{\beta}_{K_d dc} = N_{kdc} + \tilde{\beta}_{kdc}$$

$$\boldsymbol{\beta}_{dc} \sim \text{Dir}(\tilde{\boldsymbol{\beta}}_{dc})$$

$$\tilde{\boldsymbol{\beta}}_{dc} = \{1, \dots, 1\}$$

$$p(\boldsymbol{\beta}_{dc}|\mathcal{D}) = \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})$$

$$\bar{\boldsymbol{\beta}}_{dc} = \{N_{1dc} + 1, \dots, N_{K_d dc} + 1\}$$

$$p(x_d = k | y = c, \mathcal{D}) = \frac{N_{kdc} + 1}{N_c + K_d}$$

加一平滑 (add-one smoothing)
或
拉普拉斯平滑 (Laplace smoothing)

$$\boldsymbol{\beta}_{dc} \sim \text{Dir}(\tilde{\boldsymbol{\beta}}_{dc}) = \frac{1}{B(\tilde{\boldsymbol{\beta}}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\tilde{\beta}_{kdc}-1} = \frac{1}{B(\{1, \dots, 1\})}$$

→ $\tilde{\boldsymbol{\beta}}_{dc}$ 的先验分布是均匀分布

未记录到湿度值为“低”的天气情况

日期ID	天气	温度	湿度	风力	爬山
1	晴	热	高	弱	否
2	晴	热	高	强	否
3	阴	热	高	弱	是
4	雨	适中	高	弱	是
5	雨	冷	中	弱	是
6	雨	冷	中	强	否
7	阴	冷	中	强	是
8	晴	适中	中	弱	否
9	晴	冷	中	弱	是
10	雨	适中	中	弱	是
11	晴	适中	中	强	是
12	阴	适中	中	强	是
13	阴	热	中	弱	是
14	阴	冷	中	强	是
15	雨	适中	高	强	否

假定先验分布都是均匀分布

注意：这里 $\mathcal{D}$ 代表左边的数据集，
类标签1代表是，类标签2代表否

$$p(y = c|\mathcal{D}) = \frac{N_c + 1}{\sum_{i=1}^C N_i + C}$$

$$p(\text{爬山} = \text{是}|\mathcal{D}) = \frac{N_{\text{是}} + 1}{N_{\text{是}} + N_{\text{否}} + 2} = \frac{10 + 1}{10 + 5 + 2}$$

$$p(\text{爬山} = \text{否}|\mathcal{D}) = \frac{N_{\text{否}} + 1}{N_{\text{是}} + N_{\text{否}} + 2} = \frac{5 + 1}{10 + 5 + 2}$$

日期ID	天气	温度	湿度	风力	爬山
1	晴	热	高	弱	否
2	晴	热	高	强	否
3	阴	热	高	弱	是
4	雨	适中	高	弱	是
5	雨	冷	中	弱	是
6	雨	冷	中	强	否
7	阴	冷	中	强	是
8	晴	适中	高	弱	否
9	晴	冷	中	弱	是
10	雨	适中	中	弱	是
11	晴	适中	中	强	是
12	阴	适中	高	强	是
13	阴	热	中	弱	是
14	阴	冷	高	强	是
15	雨	适中	高	强	否

$$p(x_d = k | y = c, \mathcal{D}) = \frac{N_{kdc} + 1}{N_c + K_d}$$

$$N_{\text{是}} = 10, K_{\text{天气}} = 3, N_{\text{雨, 天气, 是}} = 3$$

$$p(\text{天气} = \text{雨} | \text{爬山} = \text{是}, \mathcal{D}) = \frac{3 + 1}{10 + 3} = \frac{4}{13}$$

$$N_{\text{是}} = 5, K_{\text{天气}} = 3, N_{\text{阴, 天气, 否}} = 0$$

$$p(\text{天气} = \text{阴} | \text{爬山} = \text{否}) = \frac{0 + 1}{5 + 3} = \frac{1}{8}$$

$$N_{\text{是}} = 10, K_{\text{湿度}} = 3, N_{\text{低, 湿度, 是}} = 0$$

$$p(\text{湿度} = \text{低} | \text{爬山} = \text{是}) = \frac{0 + 1}{10 + 3} = \frac{1}{13}$$

不同于贝叶斯方法通过求期望来对类标签的后验概率进行估计，最大后验估计 (MAP) 用参数后验分布概率最大值作为朴素贝叶斯模型参数的值，进而进行预测

$$p(\boldsymbol{\theta}|\mathcal{D}) = \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})$$

$$\boldsymbol{\theta}^{MAP} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\boldsymbol{\theta}|\mathcal{D})$$

$$= \underset{\boldsymbol{\alpha}, \boldsymbol{\beta}_{11}, \dots, \boldsymbol{\beta}_{DC}}{\operatorname{argmax}} \text{Dir}(\boldsymbol{\alpha}; \bar{\boldsymbol{\alpha}}) \prod_{c=1}^C \prod_{d=1}^D \text{Dir}(\boldsymbol{\beta}_{dc}; \bar{\boldsymbol{\beta}}_{dc})$$

$$\theta^{MAP} = \underset{\alpha, \beta_{11}, \dots, \beta_{DC}}{\operatorname{argmax}} \operatorname{Dir}(\alpha; \bar{\alpha}) \prod_{c=1}^C \prod_{d=1}^D \operatorname{Dir}(\beta_{dc}; \bar{\beta}_{dc})$$



$$\alpha^{MAP} = \underset{\alpha}{\operatorname{argmax}} \operatorname{Dir}(\alpha; \bar{\alpha})$$

$$\operatorname{Dir}(\alpha; \bar{\alpha}) = \frac{1}{B(\bar{\alpha})} \prod_{c=1}^C (\alpha_c)^{\bar{\alpha}_c - 1}$$

$$\bar{\alpha} = \{\bar{\alpha}_1, \dots, \bar{\alpha}_C\}$$

$$\bar{\alpha}_c = N_c + \tilde{\alpha}_c$$

$$\alpha_c^{MAP} = \frac{\bar{\alpha}_c - 1}{\sum_{i=1}^C \bar{\alpha}_i - C} = \frac{N_c + \tilde{\alpha}_c - 1}{N + \sum_{i=1}^C \tilde{\alpha}_i - C}$$

$$\beta_{dc}^{MAP} = \underset{\beta_{dc}}{\operatorname{argmax}} \prod_{c=1}^C \prod_{d=1}^D \operatorname{Dir}(\beta_{dc}; \bar{\beta}_{dc})$$

$$\operatorname{Dir}(\beta_{dc}; \bar{\beta}_{dc}) = \frac{1}{B(\bar{\beta}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\bar{\beta}_{kdc} - 1}$$

$$\bar{\beta}_{dc} = \{\bar{\beta}_{1dc}, \dots, \bar{\beta}_{K_d dc}\}$$

$$\bar{\beta}_{K_d dc} = N_{kdc} + \tilde{\beta}_{kdc}$$

$$\beta_{kdc}^{MAP} = \frac{\bar{\beta}_{kdc} - 1}{\sum_{i=1}^{K_d} \bar{\beta}_{kdc} - K_d} = \frac{N_{kdc} + \tilde{\beta}_{kdc} - 1}{N_c + \sum_{i=1}^{K_d} \tilde{\beta}_{idc} - K_d}$$

α 的先验分布为均匀分布, 即

$$Dir(\alpha; \tilde{\alpha}) = \frac{1}{B(\tilde{\alpha})} \prod_{c=1}^C (\alpha_c)^{\tilde{\alpha}_c - 1} = \frac{1}{B(\{1, \dots, 1\})} \quad \tilde{\alpha} = \{1, \dots, 1\}$$

$$\alpha_c^{MAP} = \frac{\bar{\alpha}_c - 1}{\sum_{i=1}^C \bar{\alpha}_i - C} = \frac{N_c + \tilde{\alpha}_c - 1}{\sum_{i=1}^C (N_i + \tilde{\alpha}_i) - C} = \frac{N_c}{N}$$

同最大似然
(MLE) 估计

$$\beta_{kdc}^{MAP} = \frac{\bar{\beta}_{kdc} - 1}{\sum_{i=1}^{K_d} \bar{\beta}_{kdc} - K_d} = \frac{N_{kdc} + \tilde{\beta}_{kdc} - 1}{\sum_{i=1}^{K_d} (N_{idc} + \tilde{\beta}_{idc}) - K_d} = \frac{N_{kdc}}{N_c}$$

$\tilde{\beta}_{dc}$ 的先验分布为均匀分布, 即

$$\beta_{dc} \sim Dir(\beta_{dc}; \tilde{\beta}_{dc}) = \frac{1}{B(\tilde{\beta}_{dc})} \prod_{k=1}^{K_d} (\beta_{kdc})^{\tilde{\beta}_{kdc} - 1} = \frac{1}{B(\{1, \dots, 1\})} \quad \tilde{\beta}_{dc} = \{1, \dots, 1\}$$

- 朴素贝叶斯应用于文本分类的有两个经典模型
 - 多元伯努利模型(Multi-variate Bernoulli Model)
 - 多项模型(Multinomial Model)
 - 文献 “A Comparison of Event Models for Naive Bayes Text Classification”

选读，非必须

高斯判别分析